

Chapter 1

Math

1.1 Mathematical Preliminaries

1.1.1 Continuity of Derivatives

Several results will make use of the **Schwarz theorem**, which states that if function $f : \mathbb{R}^2 \mapsto \mathbb{R}$ has continuous second partial derivatives at a given point $\mathbf{p} \in \mathbb{R}^2 = \{p_1, p_2\}^T$,¹ then

$$\frac{\partial^2 f(\mathbf{p})}{\partial x_i \partial x_k} = \frac{\partial^2 f(\mathbf{p})}{\partial x_k \partial x_i} \quad (\text{Commutative}) \quad (1.1)$$

The Schwarz theorem establishes that the result of differentiation of a function of two variables is independent of the order of that differentiation.

1.1.2 Vector Spaces

As defined by Gurtin,²

- LIN= the set of all tensors
- LIN⁺= the set of all tensors $\mathbf{S}$ with $\det \mathbf{S} > 0$
- SYM= set of all symmetric tensors
- SKW= set of all skew tensors

¹This relationship can be extended to general dimensionality of $f : \mathbb{R}^n \mapsto \mathbb{R}$, but this higher dimensionality is not used in the development that follows.

²Gurtin ME. An introduction to continuum mechanics. Academic press; 1982, Vol. 158, Jan 12.

- PSYM= set of all positive definite, symmetric tensors
- ORTH= set of all orthogonal tensors
- ORTH⁺= set of all rotations

As defined by Curnier,³ for all second-order tensors $\mathbf{S} \in \text{LIN}$, the **tracor** $\mathbb{T}$ and **deviator** $\mathbb{D}$ are fourth-order tensors are defined as

$$\mathbb{T} \triangleq \frac{1}{3} \delta_{ij} \delta_{kl} \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l \quad (1.2)$$

such that

$$\mathbb{T} : \mathbf{S} = \mathbf{S}_{\text{VOL}} \triangleq \frac{1}{3} \text{tr}(\mathbf{S}) \mathbf{1}, \quad (1.3)$$

and

$$\mathbb{D} \triangleq \left(\frac{1}{2} \delta_{ik} \delta_{jl} + \frac{1}{2} \delta_{il} \delta_{jk} - \frac{1}{3} \delta_{ij} \delta_{kl} \right) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l \quad (1.4)$$

such that

$$\mathbb{D} : \mathbf{S} = \mathbf{S}_{\text{DEV}} \triangleq \mathbf{S} - \frac{1}{3} \text{tr}(\mathbf{S}) \mathbf{1}. \quad (1.5)$$

Note, there are two other fourth-order tensors often used, the **symmetrizer** $\mathbb{S}$ and the **skewer** $\mathbb{W}$ defined, respectively, as

$$\mathbb{S} \triangleq \frac{1}{2} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l, \quad (1.6)$$

and

$$\mathbb{W} \triangleq \frac{1}{2} (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l. \quad (1.7)$$

1.1.3 Matrix Operations

Symmetry Operators

Let $\mathbf{a}$ be a matrix of dimension $n \times n$, with components a_{ij} , $i = 1 \dots n$, $j = 1 \dots n$. The symmetry operator $\text{SYM}(\cdot)$, on any square square matrix $\mathbf{a}$, is defined

$$\text{SYM}(\mathbf{a}) \triangleq \frac{1}{2} (\mathbf{a} + \mathbf{a}^T). \quad (1.8)$$

³Curnier A. Computational methods in solid mechanics. Springer Science & Business Media; 2012 Dec 6.

The skew-symmetry operator $\text{SKW}(\cdot)$, on any square matrix $\mathbf{a}$, is defined

$$\text{SKW}(\mathbf{a}) \triangleq \frac{1}{2}(\mathbf{a} - \mathbf{a}^\top). \quad (1.9)$$

Then *any* matrix $\mathbf{a}$ can be decomposed into symmetric and skew-symmetric parts through an additive decomposition as

$$\mathbf{a} = \underbrace{\frac{1}{2}(\mathbf{a} + \mathbf{a}^\top)}_{\text{SYM}} + \underbrace{\frac{1}{2}(\mathbf{a} - \mathbf{a}^\top)}_{\text{SKW}} = \text{SYM}(\mathbf{a}) + \text{SKW}(\mathbf{a}). \quad (1.10)$$

Let the symmetric, square matrix $\mathbf{o} = \text{SYM}(\mathbf{a})$ and the skew-symmetric, square matrix $\mathbf{x} = \text{SKW}(\mathbf{a})$. Then the matrix $\mathbf{a}$ can be written as

$$\mathbf{a} = \mathbf{o} + \mathbf{x}. \quad (1.11)$$

The symmetric matrix $\mathbf{o}$ has the property

$$\mathbf{o} = \mathbf{o}^\top \iff o_{ij} = o_{ji}. \quad (1.12)$$

Explicitly, for $\mathbf{o}$ as a (3×3) matrix

$$\begin{bmatrix} o_{11} & o_{12} & o_{13} \\ o_{21} & o_{22} & o_{23} \\ o_{31} & o_{32} & o_{33} \end{bmatrix} = \begin{bmatrix} o_{11} & o_{21} & o_{31} \\ o_{12} & o_{22} & o_{32} \\ o_{13} & o_{23} & o_{33} \end{bmatrix}, \quad (1.13)$$

and of the nine components of $\mathbf{o}$, only six are independent,

$$\mathbf{o} = \begin{bmatrix} o_{11} & o_{12} & o_{13} \\ o_{12} & o_{22} & o_{23} \\ o_{13} & o_{23} & o_{33} \end{bmatrix} = \begin{bmatrix} o_{11} & o_{12} & o_{13} \\ & o_{22} & o_{23} \\ \text{sym.} & & o_{33} \end{bmatrix}. \quad (1.14)$$

The skew-symmetric matrix $\mathbf{x}$ has the property

$$\mathbf{x} = -\mathbf{x}^\top \iff x_{ij} = -x_{ji}. \quad (1.15)$$

Explicitly, for $\mathbf{x}$ as a (3×3) matrix

$$\mathbf{x} = \begin{bmatrix} 0 & x_{12} & x_{13} \\ -x_{12} & 0 & x_{23} \\ -x_{13} & -x_{23} & 0 \end{bmatrix}, \quad (1.16)$$

and of the nine components of $\mathbf{x}$, only three are independent.

Remark 1.1.1. In some references, skew-symmetric is called *antisymmetric*. Here, we will use *skew-symmetric*, or sometimes just *skew* as an abbreviated form.

Remark 1.1.2. From spectral theory, the non-zero eigenvalues exist in pairs as purely imaginary numbers $0 \pm \lambda_k i$, $\lambda \in \mathbb{R}$, with

- $k = 1 \dots n$ for $\dim(\mathbf{x}) = (n, n)$ and n even and
- $k = 1 \dots (n - 1)/2$ for $\dim(\mathbf{x}) = (n, n)$ and n odd.

For even n dimensions, there are no zero eigenvalues. For odd n dimensions, there is exactly one zero eigenvalue. In the explicit example for $\dim(\mathbf{x}) = (3, 3)$,

$$\mathbf{\Lambda} = \text{EIG}(\mathbf{x}) = \begin{bmatrix} 0 & \lambda & 0 \\ -\lambda & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (1.17)$$

Remark 1.1.3. For the skew-symmetric matrix $\mathbf{x}$, the tensor $\mathbf{M} = \mathbf{x}^2$ is negative semi-definite,

$$\mathbf{v}^T \mathbf{M} \mathbf{v} \leq 0 \quad \forall \text{ vectors } \mathbf{v} \in \mathbb{R}^n. \quad (1.18)$$

1.1.4 Extrema

Arg Max

The **arguments of the maxima**, also known as **arg max** or **argmax**, at the values of the domain x that maximize the range $f(x)$.

Example 1.

Defined over the domain $x \in [0, 2\pi]$, the function $f(x) = \cos(x)$ has maximum values of 1 at $x = 0$ and $x = 2\pi$. This notion is written with the max as

$$\max_{x \in [0, 2\pi]} f(x) = \max_{x \in [0, 2\pi]} \cos(x) = 1. \quad (1.19)$$

The **arg max** is then

$$\arg \max_{x \in [0, 2\pi]} f(x) = \arg \max_{x \in [0, 2\pi]} \cos(x) = \{0, 2\pi\}. \quad (1.20)$$

More generally, for x defined over the entire real number line,

$$\arg \max_{x \in \mathbb{R}} f(x) = \arg \max_{x \in \mathbb{R}} \cos(x) = \{2\pi k\} \text{ for } k \in \mathbb{J}. \quad (1.21)$$

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